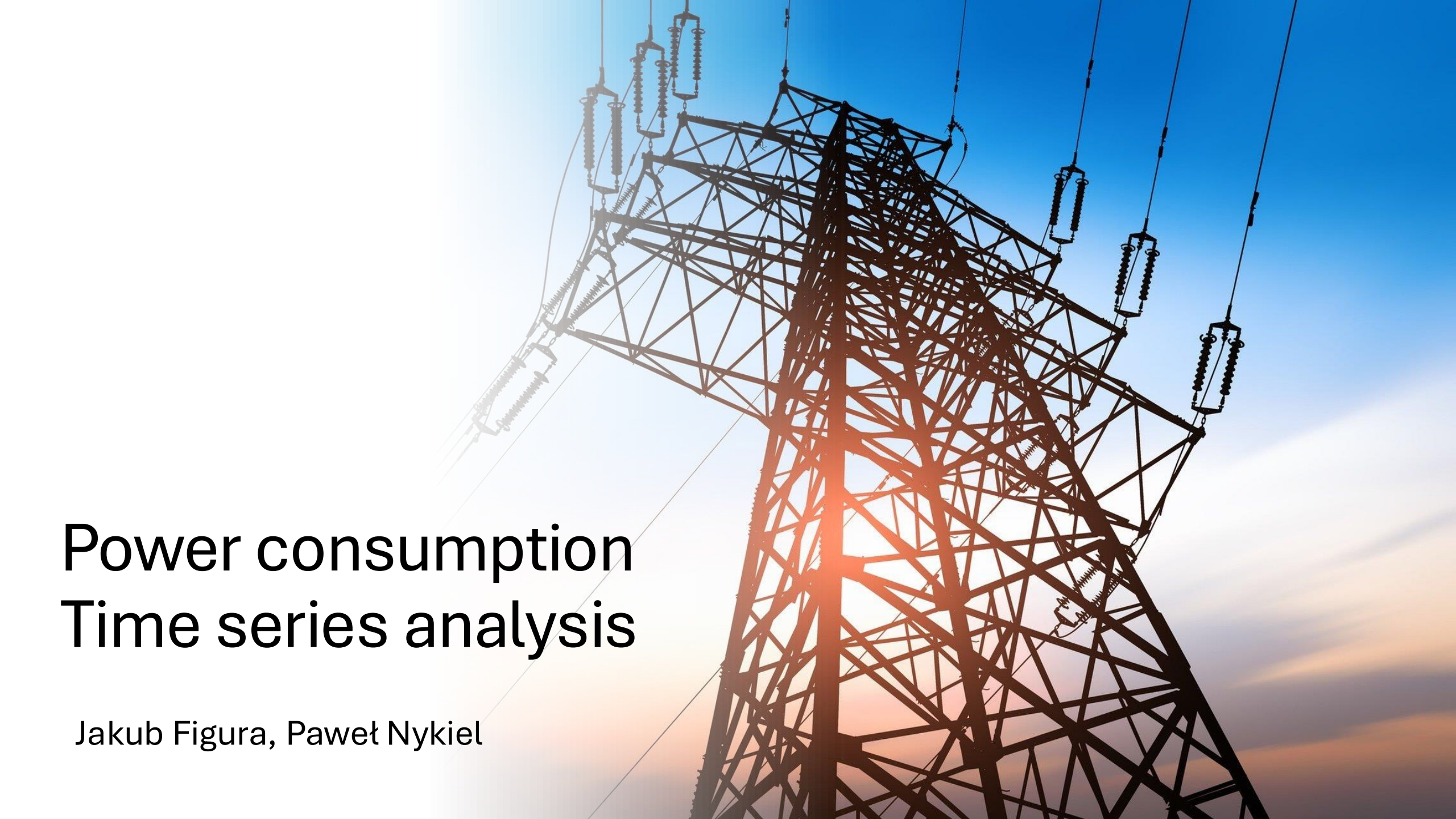




Power consumption Time series analysis

Jakub Figura, Paweł Nykiel

Problem:

Time series analysis to predict Power Consumption in the city of Tetouan Morocco

An aerial, high-angle view of a city, likely New York City, showing a dense grid of skyscrapers and buildings. The image is rendered in a semi-transparent, light blue and white style, giving it a digital or architectural feel. A semi-transparent white rectangular box is overlaid horizontally across the middle of the image, containing the text.

Three different areas of the city: **3 target variables**

Data descriptions:

- **FEATURES:**

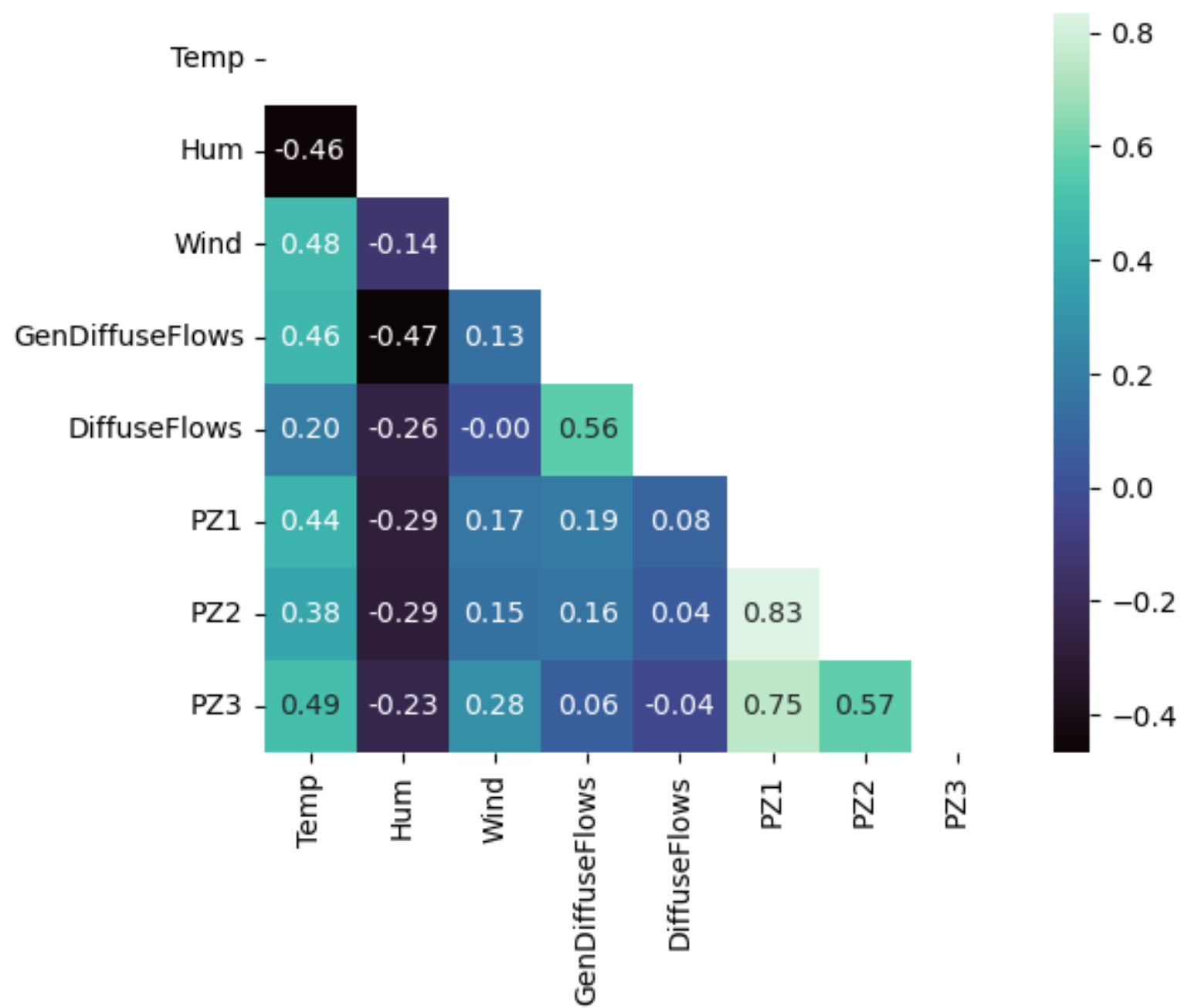
- Data Time: Time window of ten minutes [object]
- Temperature: Weather Temperature in °C [float64]
- Humidity: Weather Humidity in % [float64]
- Wind Speed: Wind Speed in km/h [float64]

Targets:

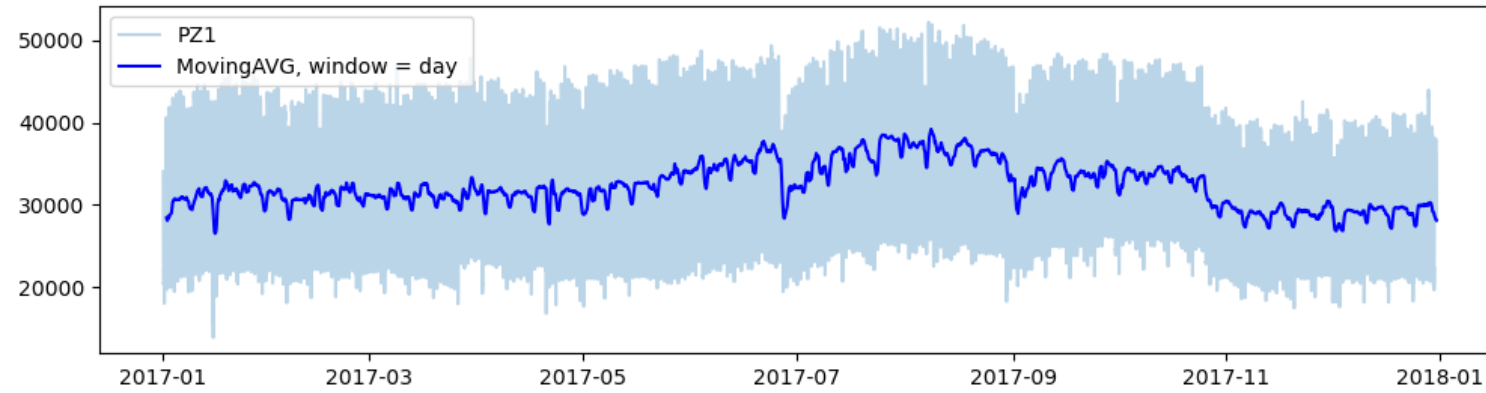
- Zone 1 Power Consumption in KW (KiloWatts) [float64]
- Zone 2 Power Consumption in KW [float64]
- Zone 3 Power Consumption in KW [float64]

- No missing values!

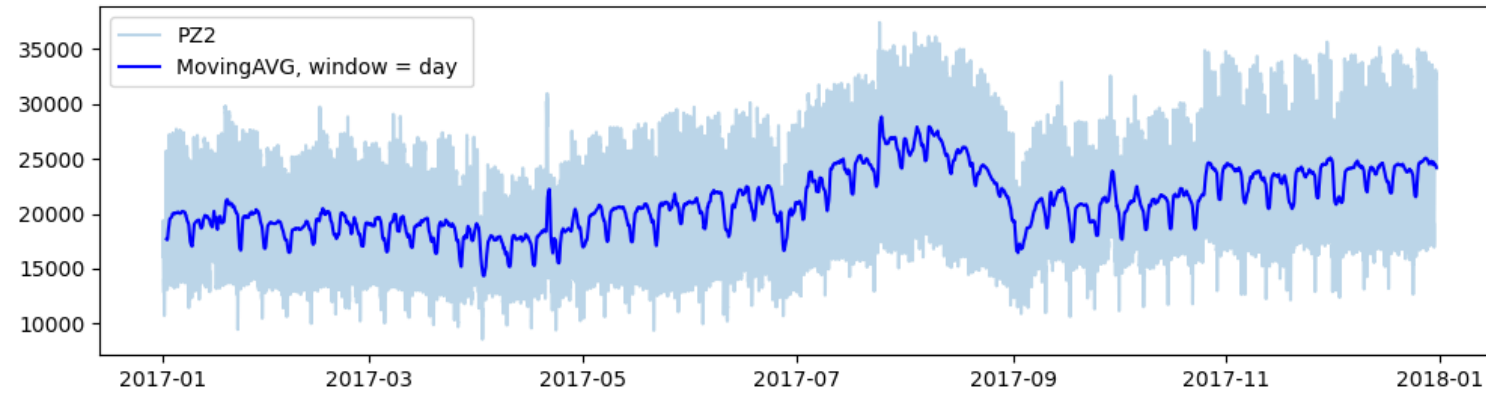
	Temperature	Humidity	Wind Speed	general diffuse flows	diffuse flows	Zone 1 Power Consumption	Zone 2 Power Consumption	Zone 3 Power Consumption
DateTime								
2017-01-01 00:00:00	6.559	73.8	0.083	0.051	0.119	34055.69620	16128.87538	20240.96386
2017-01-01 00:10:00	6.414	74.5	0.083	0.070	0.085	29814.68354	19375.07599	20131.08434
2017-01-01 00:20:00	6.313	74.5	0.080	0.062	0.100	29128.10127	19006.68693	19668.43373
2017-01-01 00:30:00	6.121	75.0	0.083	0.091	0.096	28228.86076	18361.09422	18899.27711
2017-01-01 00:40:00	5.921	75.7	0.081	0.048	0.085	27335.69620	17872.34043	18442.40964



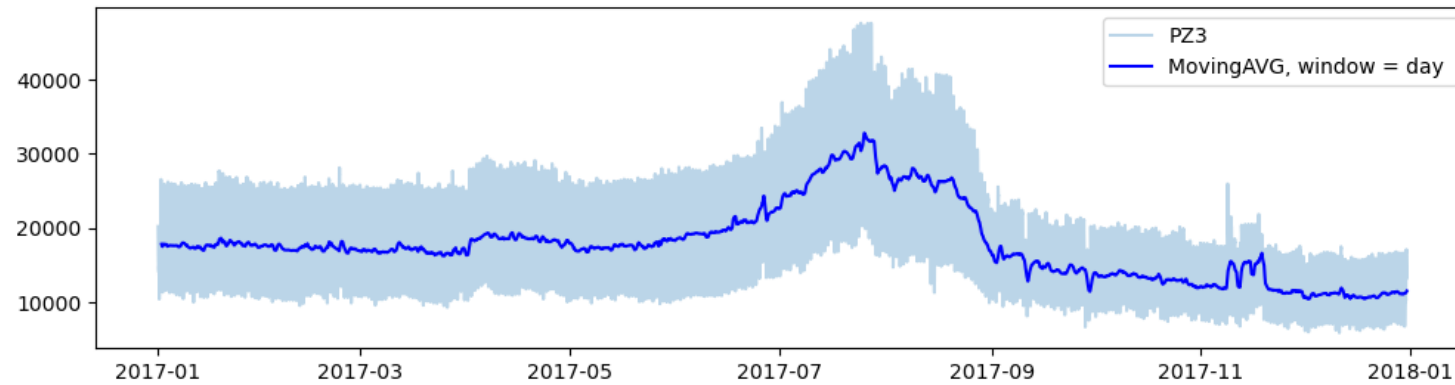
Time series for PZ1



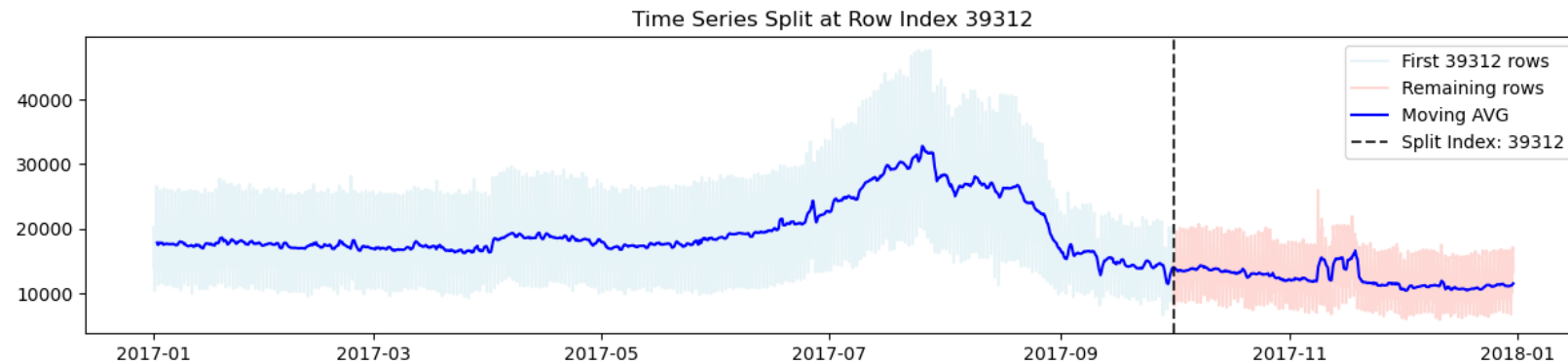
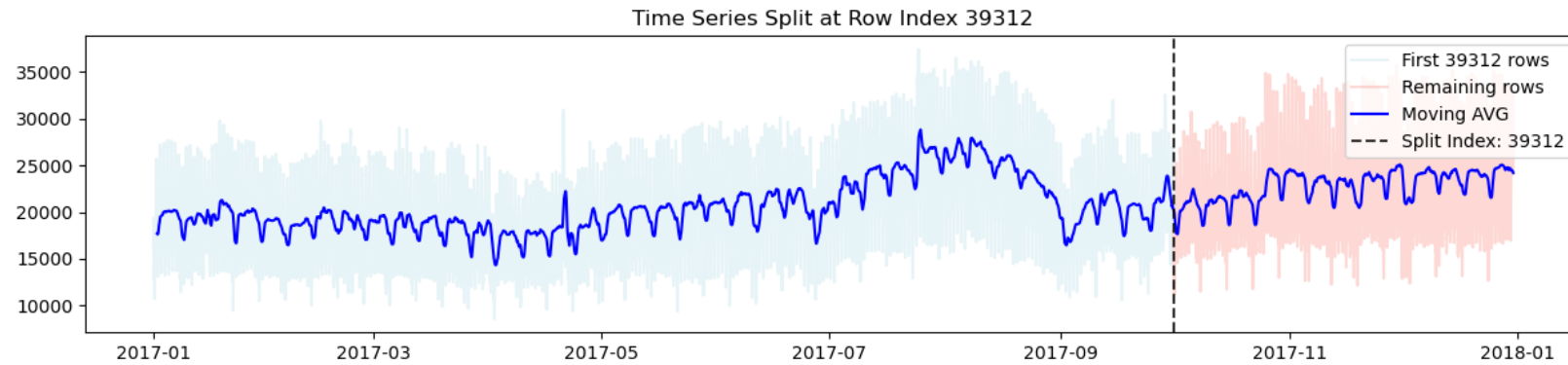
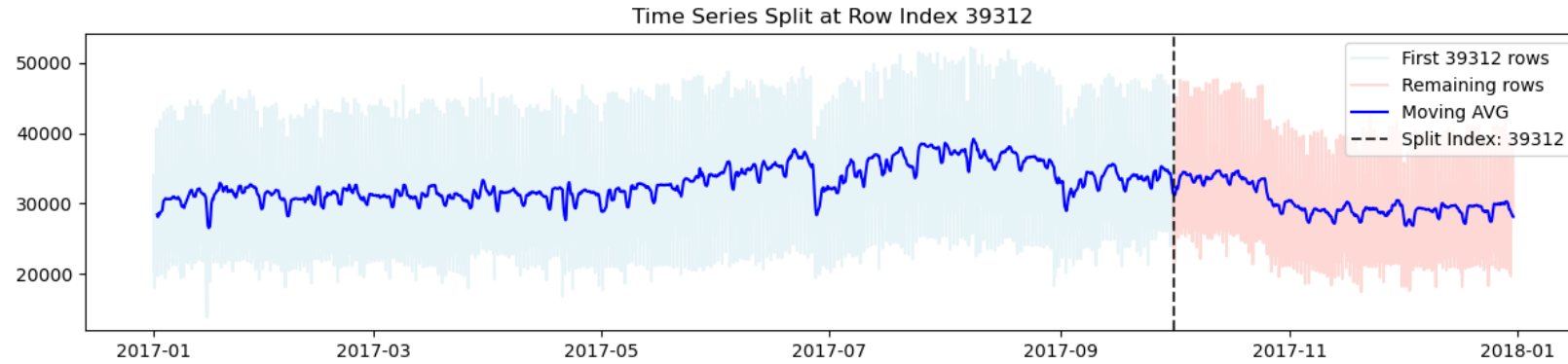
Time series for PZ2



Time series for PZ3



Test – train split: we computed index to split in ratio 75% : 25%



Models:

- Random Forest Regressor
- XGBoost
- LightGBM
- CatBoost

Results:

ZONE	MODEL	MAPE %	RMSE
PZ1	Random Forest Regressor	11.27	3600.79
PZ2	Random Forest Regressor	11.04	3622.16
PZ3	Random Forest Regressor	36.06	9434.21

ZONE	MODEL	MAPE %	RMSE
PZ1	XGBoost	12.77	4203.39
PZ2	XGBoost	9.259	2908.58
PZ3	XGBoost	14.01	2078.80

ZONE	MODEL	MAPE %	RMSE
PZ1	LightGBM	10.81	3509.80
PZ2	LightGBM	8.87	2779.10
PZ3	LightGBM	13.05	1956.54

ZONE	MODEL	MAPE %	RMSE
PZ1	CatBoost	12.9	4093.62
PZ2	CatBoost	8.29	2723.93
PZ3	CatBoost	15.69	2183.80

ZONE	MODEL	MAPE %	RMSE
PZ1	XGBoost + Optuna	8.85	3062.26
PZ2	XGBoost + Optuna	8.16	2327.65
PZ3	XGBoost + Optuna	13.82	1982.65

ZONE	MODEL	MAPE %	RMSE
PZ1	LightGBM + Optuna	7.24	2768.62
PZ2	LightGBM + Optuna	7.76	2328.77
PZ3	LightGBM + Optuna	12.59	1924.06

We set sliding window to 6, which is 1 hour.